

# The Electron Unit Charge in T0: Geometric Resolution of Point Singularities

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## Abstract

The classical representation of the electron unit charge as a point singularity encounters fundamental issues in quantum electrodynamics (QED), such as infinite self-energy and ultraviolet divergences. This treatise, authored as the creator of T0 Theory (Time-Mass Duality Framework), demonstrates how T0 resolves these singularities by treating charge as an emergent, geometric property of a universal field. Based on the single parameter  $\xi = \frac{4}{3} \times 10^{-4}$  and the Time-Mass Duality  $T_{\text{field}} \cdot E_{\text{field}} = 1$ , the charge is derived as a fractal pattern of quantized scales (effective fractal dimension  $D_f^{\text{eff}} \approx 2.973$ ). This avoids infinities, explains observations like the fine-structure constant  $\alpha \approx 1/137$ , and seamlessly connects to kinematic models in Electromagnetic Mechanics. The GitHub documentation for T0 Theory (current as of October 21, 2025) serves as a reference for detailed derivations.

## .1 Introduction: The Problem of Point Singularities

In standard physics, the electron unit charge  $-e \approx -1.602 \times 10^{-19}$  C is modeled as a Dirac delta function  $\rho(\mathbf{r}) = -e\delta(\mathbf{r})$ . This leads to a Coulomb field  $E(\mathbf{r}) \propto 1/r^2$  and infinite electrostatic self-energy:

$$U = \frac{1}{2} \int \epsilon_0 E^2 dV \rightarrow \infty \quad (\text{as } r \rightarrow 0). \quad (1)$$

QED addresses this through renormalization (vacuum polarization), yet the bare point singularity remains a mathematical artifact. Experimentally, the electron appears point-like (to  $< 10^{-22}$  m), but this does not preclude extended models at deeper scales. T0 Theory, which I developed as its creator, radically resolves this dilemma: Charge is not an intrinsic point property but an emergent projection of geometric patterns in the universal field.

## .2 Alternative Representations of Charge

### Nonlinear Electrodynamics

In models like Born-Infeld, the field saturates at maximum strength  $\beta \approx 1.2 \times 10^{20}$  V/m, yielding an effective charge radius  $r_0 = \sqrt{e/(4\pi\epsilon_0\beta)} \approx 3.5 \times 10^{-15}$  m. This results in finite self-energy  $U \approx 1.24 e^2/(4\pi\epsilon_0 r_0)$ . Here  $r_0$  is fixed by  $e/(4\pi\epsilon_0 r_0^2) = \beta$ ; with  $U = m_e c^2$  one obtains  $r_0 = 3.48 \times 10^{-15}$  m and  $\beta = 1.19 \times 10^{20}$  V/m.

### Soliton and Vortex Models

The electron as a stable wave packet in nonlinear field theories (e.g., sine-Gordon) distributes the charge density  $\rho(r)$  over a finite width, with  $E \propto q(r)/r^2$  and  $q(r) \rightarrow 0$  as  $r \rightarrow 0$ .

### Topological Defects

Charge as a Chern-Simons vortex in gauge theories, quantized by topology ( $\pi_3(S^2) = \mathbb{Z}$ ), without a bare singularity.

| Model              | Singularity?   | Self-Energy                |
|--------------------|----------------|----------------------------|
| Point Charge (QED) | Yes            | $\infty$ (renormalized)    |
| Born-Infeld        | Effectively no | Finite                     |
| Soliton            | No             | Finite (from field energy) |
| T0 Geometry        | No             | From $\xi$ -scaling        |

**Table 1:** Comparison of alternative charge representations

## .3 The Electron Charge in T0 Theory

### Time-Mass Duality and Emergence

T0 Theory unifies quantum mechanics and relativity in a parameter-free framework via  $T_{\text{field}} \cdot E_{\text{field}} = 1$ . Particles emerge as excitation patterns in the field, governed by  $\xi = \frac{4}{3} \times 10^{-4}$ . The fine-structure constant arises as:

$$\alpha = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2, \quad E_0 = 7.400 \text{ MeV}, \quad (2)$$

yielding  $\alpha \approx 7.3013 \times 10^{-3}$  ( $1/\alpha \approx 136.96$ ). With the fractally corrected  $E_0 = 7.348/\sqrt{K_{\text{frac}}} = 7.398$  MeV one obtains  $1/\alpha = 137.035$  (CODATA: 137.035999177).

The charge  $-e$  is a dimensionless geometric relation:  $q^{\text{T0}} = -1$  (in natural units), projected via  $S_{\text{T0}} = 1.782662 \times 10^{-30}$  kg onto SI values. No singularity, as the charge density is fractally distributed:

$$\rho(r) \propto \xi \cdot f_{\text{fractal}} \left( \frac{r}{\lambda_{\text{Compton}}} \right), \quad (3)$$

with  $f_{\text{fractal}}(r) = \prod_{n=1}^{137} \left( 1 + \delta_n \cdot \xi \cdot \left( \frac{4}{3} \right)^{n-1} \right)$  and effective fractal dimension  $D_f^{\text{eff}} \approx 2.973$ .

## Finite Self-Energy and Quantization

The self-energy is finite:

$$U = \frac{1}{2} \int \epsilon_0 E^2 dV = \frac{e^2}{8\pi\epsilon_0 r_e} \cdot K_{\text{frac}}, \quad (4)$$

$$r_e \approx 2.817 \times 10^{-15} \text{ m} \quad (\text{classical radius from } \xi\text{-scaling}), \quad (5)$$

$$K_{\text{frac}} = 1 - 100\xi = 74/75 \approx 0.98667 \quad (\text{fractal correction factor}). \quad (6)$$

A quantization from discrete scales of the form  $q_n = -n \cdot e \cdot \xi^{1/2}$  does not hold, because it does not yield the unit charge: with  $\xi^{1/2} = \sqrt{4/30000} = 0.011547$ ,  $n = 1$  gives the charge  $q_1 = -0.01155 e$ , not  $-e$ ;  $-e$  would arise only for  $n = \xi^{-1/2} = 86.6$ , i.e. for no integer  $n$ .

## .4 Implications for Electromagnetic Mechanics

T0 integrates with kinematic mechanics: Charge emerges as a rotating EM vortex, stabilized by fractal correction. No Dirac delta— $\rho(r)$  is a helical pattern, enabling singularity-free simulations. Applications: g-2 anomaly predictions and LHC mass spectra.

## .5 Notation

$\xi$  Geometric parameter;  $\xi = \frac{4}{3} \times 10^{-4}$

$S_{T0}$  Scaling factor;  $S_{T0} = 1.782662 \times 10^{-30} \text{ kg}$

$f_{\text{fractal}}$  Fractal function;  $\prod_{n=1}^{137} (1 + \delta_n \cdot \xi \cdot (4/3)^{n-1})$

$D_f^{\text{eff}}$  Effective fractal dimension;  $D_f^{\text{eff}} \approx 2.973$  (local:  $D_f^{\text{space}} = 3 - \xi \approx 2.9999$ )